SAMBULO ZIKHALI

Computer Science Student

ABOUT ME

A driven and academically excellent Computer Science student at the University of Pretoria, possessing a strong foundation in Object Oriented Programming, data structures, algorithms and full stack web development. From a young age, . I enjoy turning challenging business requirements into simple, well engineered solutions, and am energised by working closely with people to get there, whether that's a development team, a client or the students I mentor. Having gained enterprise software development exposure at BBD Software, as well as receiving the opportunity to broaden my commercial and analytical thinking through the Ninety One Changeblazers program, further supported by my leadership experience gained across my mentorship and committee roles, I bring strong communication skills, initiative and genuine enthusiasm for applying technology to real business problems.

CONTACT

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EDUCATION

BSC Computer Science 2024-Present
University of Pretoria

- Vice Chancellor's Distinguished Merit Award
- Dean's Merit List (2024)

National Senior Certificate (Grade 12) 2019-2023
Empangeni High School

- 7 Distinctions - 93% Average
- Top Achiever, Class of 2023
- IITPSA Olympiad Bronze Finalist

EXPERIENCE

AI Weaponised Technical Interview Platform Present
University of Pretoria, BBD Software

- Part of a development team building a technical interview platform for BBD Software designed to detect and counter AI assisted cheating during live technical interviews.
- Gaining full stack experience through a ReactJS and Tailwind CSS frontend, a FastAPI backend, integrating with a PostgreSQL database hosted on Supabase through the SQLAlchemy ORM.
- Applying industry standard software engineering practices across the full project life cycle - from planning, to testing, deployment and documentation - under the guidance of our lecturers and industry guest lecturers.

`ReactJS` `TailwindCSS` `FastAPI` `SQLAlchemy` `Supabase`

Vacation Work 2026

BBD Software

- Built and delivered a frail care facility finder under the guidance of BBD employees.
- Developed full stack features using ReactJS, NodeJS and PostgreSQL alongside Docker, gaining production level exposure to enterprise codebases.
- Navigated a full Agile software delivery lifecycle, collaborating across disciplines to translate business requirements into working software within a professional engineering environment.
- Acquired technical proficiency in backend work through direct work on APIs and server-side logic using Node.js and Express.js
- Gained exposure to dev ops using Google Firebase.
- Exposed to industry level collaboration and improved my coding standards as well as my version control skills using the Git CLI.

`ReactJS` `PostgreSQL` `Node.js` `Express.js` `Git`

University Cup Hackathon 2026

Entelect

- Competed in the Entelect University Cup, challenged to solve a complex, time pressured track race themed optimisation problem.
- Demonstrated my Java and Object Oriented Programming skills, decomposing the problem statement and applying domain modelling principles to translate the problem into software components.
- Applied critical thinking, rapid prototyping and collaborative problem solving under competitive conditions, demonstrating the adaptability and learning agility required in fast moving technical environments.

`Java` `Optimisation` `Problem Solving` `Competitive Programming`

Changeblazer 2024-Present

Ninety One

- Competitively selected for the Ninety One Changeblazers programme based on academic excellence and leadership potential, gaining direct exposure to a regulated, client facing industry and its standards for professionalism.
- Acquired practical financial and business literacy, including investment principles, asset management, and budgeting, strengthening commercial awareness alongside technical skill.
- Refined professional communication through sessions on public speaking, personal branding and operating with confidence in corporate environments.

`Finance` `Networking` `Personal Branding` `Investment Basics`

LEADERSHIP

Head of IT Present
VCDMA Alumni Chapter Committee

- Serving as Head of IT for the VAC Committee, taking ownership of technical infrastructure and digital systems supporting the committee's activities. Leveraging AI agents to boost productivity for the committee.

Mentor Present
STARS Mentorship Programme

- Mentoring students through the University of Pretoria's STARS Mentorship Programme, supporting their transition into university, as well as their academic and personal growth.

Mentor Present
TuksRes

- Mentoring fellow TuksVilllage residence occupants, providing guidance to peers navigating university life, as well as guiding them through the various facilities and support systems offered by the residence.

Prefect, Representative Council of Learners Member 2023
Empangeni High School

- Held elected leadership positions, demonstrating integrity, accountability and reliability in positions of trust.

Peer Tutor 2023
Empangeni High School

- Assisted fellow students in Mathematics, Physics, Information Technology and Computer Applications Technology, breaking down difficult concepts into digestible and understandable information.

SKILLS

Programming languages: TypeScript, Python, Java, C++, JavaScript, PHP.

Web/Frameworks: HTML/CSS, ReactJS, Angular, Next.js, Express.js, FastAPI, REST APIs.

Databases/ORMS: PostgreSQL, MySQL, Prisma, SQLAlchemy.

Tools/Practices: Git, Linux, Docker, Agile Development.

AI Proficiencies: Claude, ChatGPT, GitHub Copilot, Google Gemini Pro, Google Antigravity.

Core Competencies: Leadership, Critical thinking, Adaptability, Curiosity, Meticulous in the work I deliver.

INTERESTS

Tinkering with digital systems, Gaming, Football, Weightlifting, Tennis, Car Photography.